

IAT 461 Final Project Progress Report

Restaurant Complaint Theme Discovery

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1. Project Overview

To identify distinct and actionable complaint themes, this project looks at restaurant reviews from Turkey. It then determines whether these themes are concentrated at certain locations or recur across multiple restaurants. A stakeholder is a director of marketing and outreach who requires a succinct explanation of customer dissatisfaction instead of just raw reviews and star ratings.

This is an unsupervised text discovery problem since there are no predefined labels with a complaint theme in the dataset. A small number of interpretable themes backed by representative reviews and cross-restaurant comparisons are the desired outcome.

2. Work Completed to Date

- Reviewed the Client proposal, stakeholder need, and success criteria
- Completed an initial data quality audit covering dimensions, missingness, ratings, restaurant coverage, and duplicate text
- Classified the task as unsupervised text clustering and outlined the required EDA, modeling, interpretation, evaluation, executive summary, and limitations sections

3. Initial Dataset Profile(Self Notes)

- **Size:** 1,100 reviews and 6 columns
- **Coverage:** 100 restaurants with exactly 11 reviews per restaurant
- **Core fields:** business_name, text, rating, rating_category, author_name, and photo
- **Missingness:** No missing values were found in the supplied columns
- **Ratings:** 80 one-star, 72 two-star, 172 three-star, 316 four-star, and 460 five-star reviews
- **Complaint subset:** 324 reviews have ratings of 1-3- the final complaint threshold will be justified during EDA
- **Duplicates:** 12 exact duplicate review texts require investigation before modeling
- **Existing categories:** taste, menu, indoor atmosphere, and outdoor atmosphere. These describe review dimensions, not discovered complaint themes

4. Early Findings and Project Risks

- Restaurant-specific percentages must be displayed with counts and interpreted cautiously because the dataset is balanced by restaurant, but only 11 reviews are available for each restaurant.

- Filtering solely by star rating may overlook mixed or "hidden" complaints because some highly rated reviews still contain criticism.
- In certain places, the review wording seems to be translated or inconsistent. To maintain the meaning of negation and complaints, cleaning must be minimal.

5. Immediate Next Steps

- **EDA and cleaning:** Analyze rating distribution, review length, restaurant coverage, duplicates, and existing rating categories; preserve the original text while creating a lightly cleaned version.
- **Complaint-corpus definition:** Compare a ratings-based subset, provisionally ratings 1–3, with the full review text so criticism inside positive reviews is not overlooked.
- **Baseline model:** Represent review text with TF-IDF and apply K-means. Test several values of K using the elbow method, silhouette score, cluster size, and interpretability.
- **Cross-restaurant analysis:** Measure how widely each theme appears and identify themes that are widespread versus concentrated at a small number of restaurants.

6. Current Status

The project is presently in the data understanding and planning stage, meaning that the provided dataset has been verified and profiled, the main risks have been identified, model training and theme validation have not yet been completed and has translated into an analytical question. Completing EDA and creating the first interpretable clustering baseline is the next milestone.